



Four- to 6-month-old infants use saliva-sharing interactions to infer social intimacy[☆]

Brandon M. Woo^{a,*}, Elizabeth S. Spelke^b, Rebecca Saxe^c, Ashley J. Thomas^b

^a Department of Psychological & Brain Sciences, University of California, Santa Barbara, United States of America

^b Department of Psychology, Harvard University, Cambridge, United States of America

^c Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, United States of America

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ABSTRACT

People form distinctive relationships that are close, intimate, or “thick”. Across cultures, these relationships are often characterized by physical intimacy, including actions that involve sharing saliva, such as kissing or sharing foods or utensils. Recent research by Thomas et al. (2022) provides evidence that infants recognize saliva sharing as evidence of a thick relationship: Eight- to 10-month-old infants expect that two characters who share saliva will subsequently respond to each other's distress. To constrain theories of how knowledge of thick relationships emerges in development, the current studies tested for this expectation in younger infants, who have limited experience sharing food within the family and observing other people doing so. We tested 4- to 6-month-old infants in the same paradigms that Thomas et al. (2022) used to study 8- to 10-month-old infants. We replicated the findings from older infants in this younger sample; when 4- to 6-month-old infants saw a central actor in distress, they looked longer to an actor who had previously shared saliva with the central actor than to another actor who had not. These findings suggest that 4- to 6-month-old infants distinguish thick from other social relationships.

To become competent members of their social networks, children must learn how people are connected to one another (Fiske, 1992; Hirschfeld, 2003; Thomas, 2026; Thomsen & Carey, 2013). Relationships can vary in attachment, responsiveness, and intimacy (Clark & Mills, 1979; Fiske, 1992; Rai & Fiske, 2011; Sahlin, 2013). We refer to relationships high and low in these attributes as “thick” and “thin”, respectively (Margalit, 2023; see also Carsten, 1995; Fiske & Fiske, 2007; Granovetter, 1973; Steele & Thomas, in press). Research suggests that 8- to 10-month-old infants have intuitions about thick relationships, expecting individuals who share saliva to be more likely to respond to one another's distress than those who do not (Thomas et al., 2022). By 8 months, however, infants may have had several months of experience eating and sharing solid foods with others and observing people comforting one another. Do infants with even less experience infer thick relationships from acts of saliva sharing? We addressed this question through research on infants from 4 to 6 months of age.

1. Saliva sharing: A cue to thick relationships

Across cultures, people are more comfortable sharing saliva (e.g., kissing or licking the same food item) within intimate, or thick, relationships, including kin broadly defined (Carsten, 1995; Rai & Fiske, 2011; Sorokowska et al., 2021; Thomas et al., 2022). A willingness to share saliva is therefore a valid cue to intimacy (Billingsley et al., 2019). Indeed, US adults infer social closeness from eating that involves sharing saliva (Chen et al., 2026; Hung et al., 2022; Miller et al., 1998).

Children appear to share the intuition that people in thick relationships share saliva: When food involves sharing saliva (e.g., a shared ice cream cone), US children predict that a child will share with family over friends; by contrast, when food can be shared without exchanging saliva (e.g., separate candies), they make no systematic predictions (Thomas et al., 2022). Children also expect saliva sharing to occur in friendships marked by emotional intimacy (Richardson & Thomas, 2025), further suggesting they associate saliva sharing with thick relationships.

How do children learn about thick relationships? By late childhood, children have accumulated years of experience and may have received

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* Corresponding author at: 2831 Psychology East, University of California, Santa Barbara, CA 93106, United States of America.

E-mail address: bmwoo@ucsb.edu (B.M. Woo).

explicit instruction about which actions are acceptable in which relationships. Moreover, children's understanding of kinship terms and inheritance emerges late in development and varies across cultures (Astuti et al., 2004; Landau, 1982; Mollica & Piantadosi, 2022; Spokes & Spelke, 2016). Does a broader understanding of thick relationships likewise require years of experience?

Relative to older children, infants and toddlers have had fewer opportunities for observation and explicit instructions concerning the behaviors that are acceptable in close and more distant relationships, offering a way to ask how experience shapes this understanding. Thomas et al. (2022) presented infants (8 to 10 months) and toddlers (15 to 19 months) with scenes that either did or did not imply saliva sharing. In the saliva-sharing interaction, a woman and a puppet shared a slice of orange, each putting the same slice in their mouths. In the other, a different woman and the same puppet passed a ball back and forth, contacting it with their hands but not their mouths. When the puppet then cried in the presence of both women, infants and toddlers looked first and longer at the woman who had shared the orange with the puppet, as though they expected her to respond (see Ciftci et al., 2025, for replication). In further experiments, Thomas et al. (2022) presented infants and toddlers with a different saliva-sharing interaction that did not involve food. A woman placed her finger in her own mouth, then a puppet's mouth, and then her own again. This action is likely less common: Although infants and their parents may put their fingers in one another's mouths, infants are unlikely to observe two adults doing so with each other. This saliva-sharing interaction was compared to a scene in which the same woman touched her own forehead, then another puppet's forehead, and then her own again. Infants and toddlers expected the saliva sharing puppet to respond to the woman's distress, providing evidence that early-emerging abilities to distinguish thick from more distant social relationships are not limited to contexts involving everyday actions like the sharing of food. After observing social interactions with and without saliva-sharing, infants and toddlers inferred thick relationships connecting characters who shared saliva and used that inference to predict who would respond to their social partner's distress.

1.1. The present research

Thomas et al. (2022) provide evidence that extensive observation and explicit teaching are not necessary for humans to use saliva sharing to recognize thick relationships. Still, it remains unclear how much experience this understanding requires. One possibility is that it depends on many months of observing how people in thick relationships interact, on first-person experience of eating with others, or both. By 8 months of age, most infants are eating solid foods and may have several months of experience sharing saliva with others and observing others' saliva-sharing interactions. In other domains, first-person experience has been shown to affect infants' expectations (e.g., their expectations of

others' actions; Sommerville et al., 2005) and it may likewise shape how they understand saliva sharing interactions. A recently proposed connectionist model of the role of experience in infants' understanding of saliva-sharing predicts that infants will associate saliva sharing with responses to distress only with sufficient experience (Benton & Hoch, 2026). If this model is correct, at minimum, younger infants may show weaker effects, or, they may fail to show the effect at all (Benton & Hoch, 2026).

Alternatively, representations of thick social relationships may be part of infants' core knowledge (Spelke, 2022; Thomas, 2026). If so, even very young infants may be sensitive to the closeness of the social relationships that people form with others and infer that different types of relationships are likely to elicit different actions (Thomas, 2026; see also Samani & Thomas, 2026). Some cues are cross-culturally prevalent, including affective touch, bodily synchrony, and saliva sharing (Fiske, 1992; Sorokowska et al., 2021; Thomas et al., 2022). If infants have very early developing or innate knowledge of different types of social relationships, that knowledge may make certain social cues easy to detect, and certain social relationships easy to learn about and to track across contexts.

The present experiments speak to these possibilities by testing 4- to 6-month-old infants. Relative to 8- to 10-month-old infants, 4- to 6-month-old infants have been alive for roughly half as long. However, their actual experience, watching others eat and exchanging saliva (e.g., by kissing), is even lower because younger infants sleep for longer periods (Dias et al., 2018; Iglowstein et al., 2003), have lower visual acuity (Dobson & Teller, 1978), are less likely to look at faces because they cannot sit up (Frank et al., 2013), and are more likely to ingest liquid foods (e.g., breast milk or formula, foods that are typically not shared between family members; Barrera et al., 2018; Meek et al., 2022). Barrera et al. (2018) analyzed data from 2009 to 2014 for the US National Health and Nutrition Examination Survey (including data from 1482 children), and reported (i) that solids are introduced to 16.3% of infants before they turn 4 months of age and 38.3% of infants between 4 and 6 months of age; and (ii) that 91.8% of children have been introduced to solid foods by 8 months of age. Thus, over 90% 8- to 10-month-old infants have been introduced to solid foods, compared to around half of 4- to 6-month-old infants. These are large differences in relative experience. Relative to older 8- to 10-month-old infants, then, younger infants have had substantially fewer opportunities to observe and participate in saliva-sharing interactions themselves. By testing 4- to 6-month-old infants, we aimed to constrain theories of how humans come to understand thick relationships.

We applied paradigms developed by Thomas et al. (2022), which build on findings in nonhuman primates: When vervet monkeys hear a juvenile vocalize distress, they look toward that juvenile's mother, as though anticipating her response (Seyfarth & Cheney, 2003). We examined whether 4- to 6-month-old infants expect that when someone is distressed, a thick relationship partner is likely to respond (Fig. 1).

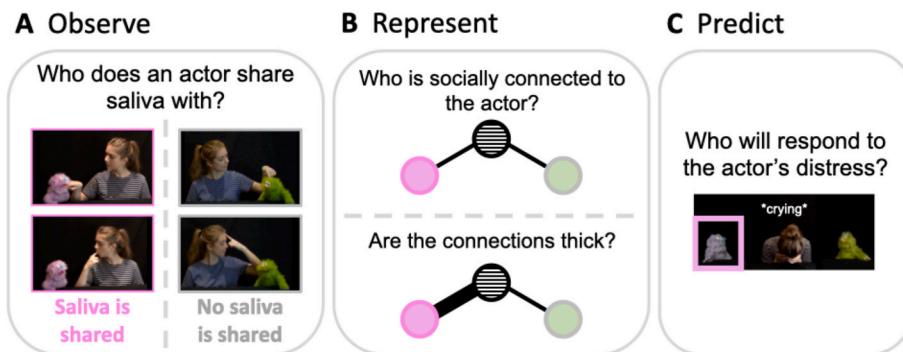


Fig. 1. A conceptual schematic illustrating the logic of the studies. (A) Infants observe a central actor who shares saliva with one actor and does not share saliva with another actor. (B) From such observations, infants may infer and represent whether the actors are connected and how thick each connection is. (C) These representations support infants' expectations of who will respond to the central actor's distress. This figure is adapted from Thomas, 2026.

2. General methods

We conducted two studies, each testing whether 4- to 6-month-old infants expect that social partners who share saliva will respond to one another's distress. In Study 1, infants watched two actors interact with a central puppet: one shared a single orange slice with it (a saliva-sharing act), and the other passed a ball back and forth with it (no saliva sharing; Fig. 2A and B). The puppet then expressed distress, and we measured whom infants looked to. We reasoned that if infants infer thick relationships from sharing saliva—even with limited exposure to it—they should look longer at the actor who had shared the orange slice.

Study 2 depicted saliva sharing without food sharing: an actor placed her finger in her own mouth, then in a puppet's mouth, then back in her own mouth. The contrasting scene was matched but involved no saliva—she first touched her own forehead, then a puppet's forehead, then her own forehead. Study 2 extended Study 1 in three ways. First, it asked whether infants expect a response to distress when no food is exchanged. Second, Study 2 asked whether infants expect mutual responsiveness in saliva-sharing interactions; whereas the individual who initiated the saliva-sharing interaction was in the position to respond in Study 1, the recipient of the saliva-sharing interaction was the one who was in the position to respond in Study 2. Lastly, Study 2 asked whether infants make inferences about thick relationships even for saliva-sharing actions that they have had less experience with; infants have likely had less experience observing people put their fingers in each other's mouths, relative to their experience observing people share food.

Studies 1 and 2 each had three parts. The main test experiments (Studies 1A and 2A) asked our central question: Do infants expect

individuals who share saliva to respond to one another's distress? Studies 1B and 2B sought to replicate the findings of Studies 1A and 2A (in a test condition) and investigated (in a control condition) whether infants expect saliva-sharers to respond to anyone's distress, or only the specific individual they had shared saliva with. Finally, Studies 1C and 2C served as control experiments, asking whether infants expect saliva-sharing partners to respond to a social partner's behavior broadly, or specifically to distress.

2.1. Open science practices

The hypotheses, methods, and analysis plans were preregistered prior to data collection: <https://osf.io/dsa52/overview><https://osf.io/dsa52/overview>. There were deviations from the preregistrations, summarized below; due to assumptions for standard *t*-tests not being met, we instead used Wilcoxon tests (see “General Methods”). Materials, data, and analysis scripts are publicly available: <https://osf.io/dsa52/overview>. We note further deviations below.

2.2. Participants

In Studies 1A and 2A, forty-six 4- to 6-month-old infants contributed data (26 girls, 20 boys; mean age = 5.39 months, range = 4;10–6;19). Thirty of these infants completed both Studies 1A and Study 2A, 12 infants only contributed data to Study 1A, and four infants only contributed data to Study 2A. Five infants began Study 1A but were excluded due to failing to make an anticipatory look ($n = 2$), fussiness ($n = 1$), and equipment failure on the infants' side ($n = 2$). Thirteen infants began Study 2A but were excluded due to failing to make an anticipatory

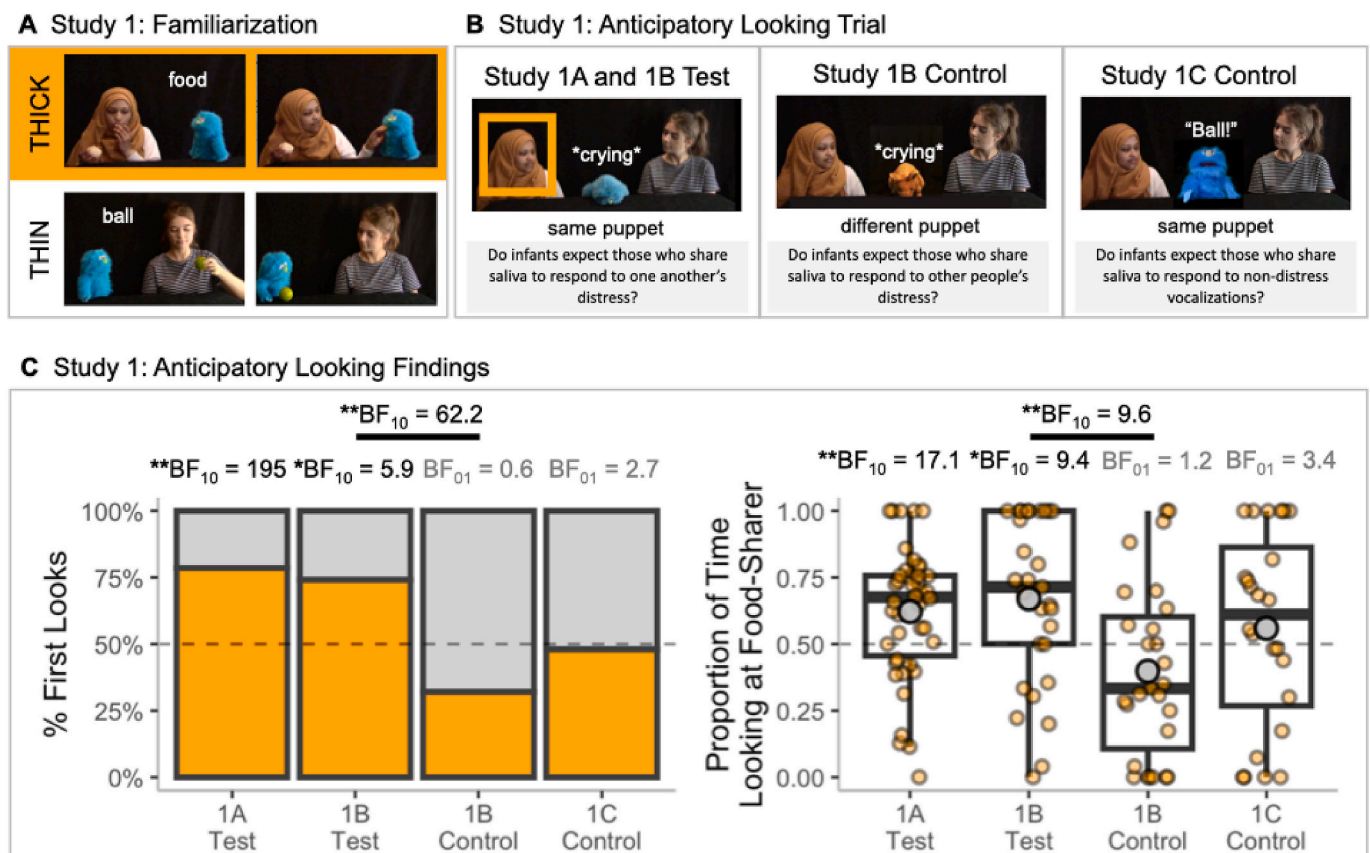


Fig. 2. Study 1's (A) familiarization events, (B) anticipatory looking trial, and (C) results: the percentage of infants who looked first toward the food sharer (the thick relationship partner) or the ball passer (the thin relationship partner) and the proportion of time infants looked toward the food sharer. Gray circles indicate means, and thick black lines indicate medians. Asterisks indicate Bayes Factors (**BF₁₀ > 10, *BF₁₀ > 3). Bayes factors for evidence in favor of H₁ over H₀ are in black, whereas Bayes factors for evidence in favor of H₀ over H₁ are in gray.

look ($n = 8$), poor video quality ($n = 2$), experimenter error ($n = 2$), and the infant having covered their eyes during the test trial ($n = 1$). We asked participants to complete demographics forms. In Studies 1A and 2A, 12 of 46 participants' parents answered questions about their children's racial and ethnic identity. Nine parents said their child was White and not Hispanic or Latinx; two said that their child was Black and not Hispanic or Latinx; one said that their child was Asian and not Hispanic or Latinx; and the rest did not report on their child's racial and ethnic identities.

In Studies 1B and 2B, sixty 4- to 6-month-old infants contributed data (31 girls, 29 boys; mean age = 5.35 months, range = 4;5–7;2). Infants were randomly assigned to participate in either (i) the control condition of Study 1B (food-sharing interaction) and the test condition of Study 2B (finger-in-mouths interaction) or (ii) the test condition of Study 1B and the control condition of Study 2B. Forty-nine of the infants contributed data to both Studies 1B and 2B, three infants only contributed data to Study 1B, and eight infants only contributed data to Study 2B. Thirteen more infants began Study 1B but were excluded due to equipment failure ($n = 7$), failure to make an anticipatory look ($n = 2$), inattentiveness ($n = 2$), and parental or sibling interference ($n = 2$). Twelve infants began Study 2B but were excluded due to poor video quality ($n = 5$), failure to make an anticipatory look ($n = 4$), inattentiveness ($n = 2$), and sibling interference ($n = 1$). In Studies 1B and 2B, 17 of 60 participants' parents answered questions about their children's racial and ethnic identity. Twelve parents said their child was White and not Hispanic/Latinx; two said that their child was Asian and not Hispanic/Latinx; one said that their child was Black or African American and not Hispanic/Latinx; one said that their children were multiracial and not Hispanic/Latinx; one responded "other"; and the rest did not report on their children's racial and ethnic identities.

In Studies 1C and 2C, twenty-six 4- to 6-month-old infants contributed data (12 girls, 14 boys; mean age = 5.31 months, range = 4;6–6;29). Twenty of these infants contributed data to both Studies 1C and 2C, five of these infants only contributed data to Study 1C, and one of these infants only contributed data to Study 2C. Seven more infants began Study 1C but were excluded due to poor video quality ($n = 4$), failure to make an anticipatory look ($n = 2$), and fussiness ($n = 1$). Nine infants began Study 2C but were excluded due to poor video quality ($n = 4$), fussiness ($n = 2$), failure to make an anticipatory look ($n = 2$) and inattentiveness ($n = 1$). In Studies 1C and 2C, due to researcher error, we did not ask about racial or ethnic information in our demographic forms.

2.3. Procedure

Each participating family met with an experimenter via a video call using the program Zoom. Infants sat on their caregiver's lap or in a highchair. Caregivers were asked to remain neutral and refrain from reacting to the videos; all caregivers complied with this request.

The typical procedure was as follows: During the video call, the experimenter shared a link to a YouTube playlist that featured an informational video, a video to test that the participant's audio was working, a calibration video, the familiarization videos, and the test trials. After the parent clicked on the YouTube playlist, they were asked to share their screen with the experimenter. This step ensured that the recording would capture both the participant's face and the stimuli they saw. Occasionally, this process was not possible because some tablet-style computers do not allow simultaneous screen sharing and video recording. In these cases, the experimenter shared their screen with the parent and played the YouTube videos. Participants tested under these viewing conditions were included in the analysis. We followed best practices for testing infants online (refer to [Chuey et al., 2021](#)).

The calibration video featured rotating stars accompanied by a high-pitched noise to capture the participants' attention. The stars moved to different positions on the screen, allowing us to have a video recording of the participant looking at various parts of the screen, which aided coding.

2.3.1. Familiarization

Infants were shown eight familiarization videos in Study 1 and six in Study 2. Each video showed three actors: one on the left, one on the right, and one in the center. We refer to the middle actor as the central actor; this actor interacted with the other two. Half of the familiarization videos involved the central actor engaging in saliva sharing with one of the other actors, and the other half involved the central actor engaging in social actions that did not involve saliva sharing with the other actor. In Study 1, the central actor was a puppet, who engaged in saliva sharing with one human actor by putting the same orange slice in both of their mouths. Another human actor passed a ball back and forth with the same puppet. In Study 2, the central actor was a human. In the saliva-sharing interaction she placed her finger in her own mouth and then a puppet's mouth; with another puppet, she placed her finger on her own forehead and then the puppet's forehead.

In both studies, both actions were social, but only one action per study involved saliva exchange. The saliva-sharing and non-saliva sharing actions were presented in alternation, with an attention-grabbing video played in between to reorient the infant.

2.3.2. Test Trials

After the familiarization videos, infants saw the test trials ([Figs. 2B and 3B](#)), in which the central actor demonstrated distress. We measured anticipatory looking for 7 s. In Studies 1A, 1B, and 2B, due to experimenter errors, there were inconsistencies in trial duration that required supplementary analyses; see supplemental materials.

2.4. Design

Each infant participated in one condition of both Study 1 and 2 in a single session. Infants participated in either Studies 1A and 2A; the test condition of either 1B or 2B and the control condition of Study 1B or 2B (i.e., the test condition in one study and the control condition in the other), or 1C and 2C.

2.5. Counterbalancing

The two actors in Study 1 and the two puppets in Study 2 were always on the same sides of the screen. We counterbalanced the identity of the actor or puppet who engaged in the thick or thin interaction, thereby also counterbalancing the side of the thick relationship partner. The order in which infants participated in Studies 1 and 2 was counterbalanced.

2.6. Coding, Analysis Plan, Sample Size Justification, and Stopping Rule

2.6.1. Coding

Coding was done offline by coders blind to the identity of the thick and thin relationship partners. Using iMovie (as in [Thomas et al., 2022](#)), coders recorded the direction of infants' first looks immediately following the central character's expression of distress and measured the duration of infants' looking during this window. The primary dependent measures were the direction of infants' first looks (toward the individual to the left or right) and the proportion of time that infants spent looking at either individual during the pause following the distress.

Across all studies, we followed [Thomas et al.'s \(2022\)](#) criterion for recording first looks. This criterion was preregistered for Studies 1C and 2C, but not for Studies 1A, 1B, 2A, and 2B. If an infant was already looking at one of the individuals when the coding window started, we counted that look as the infant's first look only if it lasted at least 0.4 s; otherwise, we coded their next look as their first. Fifteen infants did not meet the criterion of 0.4 s (4 in Study 1A; 5 in Study 2A; 3 in Study 1B; 2 in Study 1C; 1 in Study 2C); we therefore coded their next look as their first.

We also calculated the proportion of looking, as in [Thomas et al. \(2022\)](#). We calculated each infant's proportion of looking at the thick

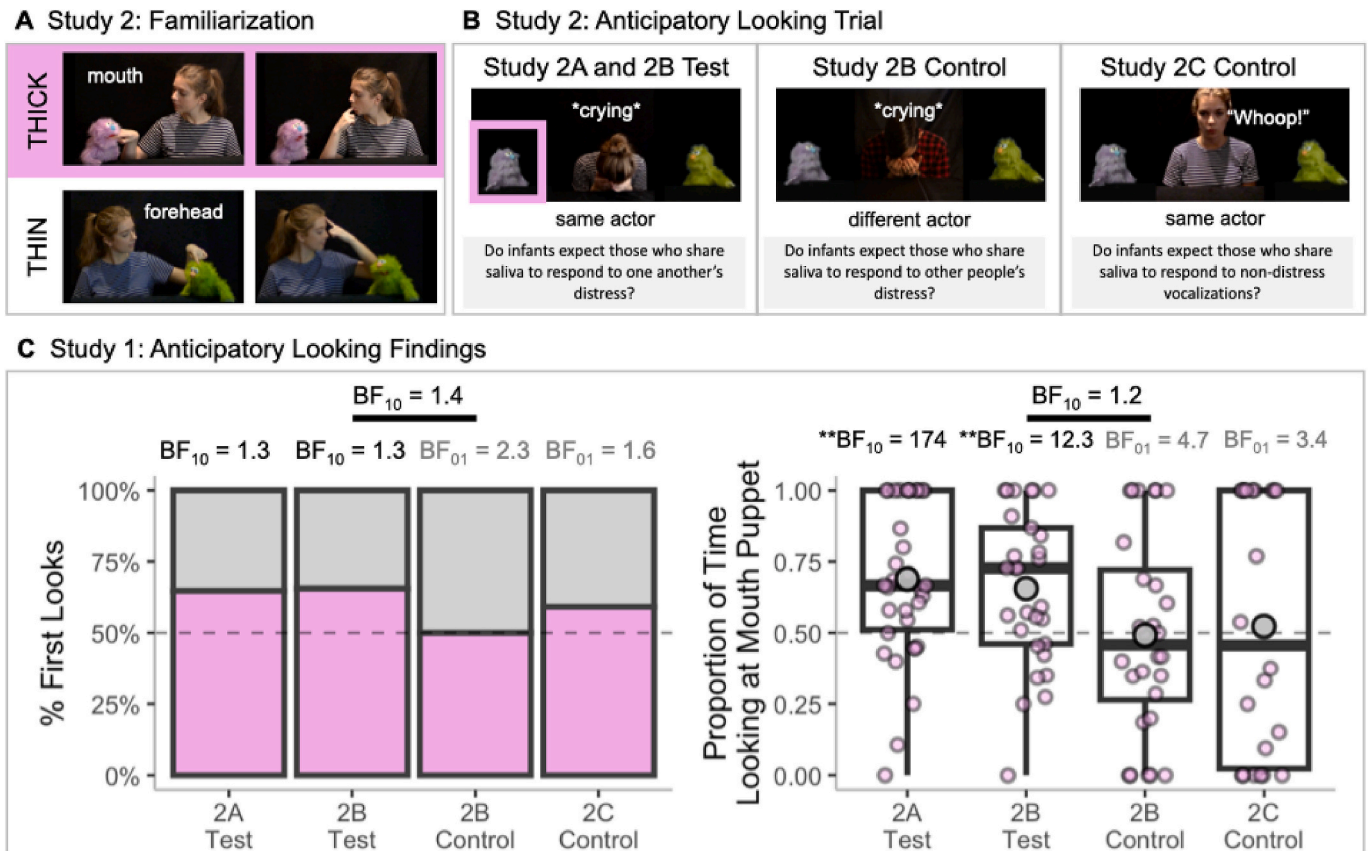


Fig. 3. Study 2's (A) familiarization events, (B) anticipatory looking trial, and (C) results: the percentage of infants who looked first toward the mouth puppet or the forehead puppet, and the proportion of time infants looked toward the mouth puppet. Gray circles indicate means, and thick black lines indicate medians. Asterisks indicate Bayes Factors (**BF₁₀ > 10, *BF₁₀ > 3). Bayes factors for evidence in favor of H₁ over H₀ are in black, whereas Bayes factors for evidence in favor of H₀ over H₁ are in gray.

relationship partner by dividing time spent looking at the thick relationship partner by total time looking at either partner.

2.6.2. Inter-coder reliability

We assessed reliability by double-coding 22 participants' trials in Studies 1A, 1B, 2A, and 2B, and 14 different participants' trials in Study 1C and 2C. We had preregistered double-coding 10% of trials, but the research assistant happened to double-code additional participants. We included all double-coded data in our analyses for a better estimate of reliability, and we did not inspect the data until all the data had been double-coded. Because there were two sets of coders, we calculated reliability separately for the two sets. For Studies 1A, 1B, 2A, and 2B, the correlation coefficient between the two coders' looking times was 0.92 (Krippendorff's $\alpha = 0.74$), and the percentage of agreement between their first look judgments was 90% (Cohen's $\kappa = 0.80$). For Studies 1C and 2C, the correlation coefficient between the two coders' looking times was 0.94 (Krippendorff's $\alpha = 0.85$), and the percentage of agreement between their first look judgments was 91% (Cohen's $\kappa = 0.79$).

2.6.3. Sample size and stopping rule

Across studies, we used Bayes factors to weigh the evidence for the alternative and null hypotheses. We planned to stop data collection upon obtaining a Bayes factor of at least 8 (Kass & Raftery, 1995). Following Thomas et al., 2022, we preregistered testing at least 18 infants before checking Bayes factors in Studies 1A and 2A, and at least 24 infants in Studies 1B and 2B. The larger sample size in Studies 1B and 2B reflects that there were two conditions, for which we wanted to test for differences between conditions. For Studies 1C and 2C, we did not preregister a minimum sample size, but we followed the plan for Studies 1A and 2A:

testing 18 infants and then checking the Bayes factors.

If a Bayes factor of 8 was not obtained at the minimum sample size, we planned to continue testing until reaching it or a preregistered maximum of 36 infants in Studies 1A and 2A, 80 infants in Studies 1B and 2B, and 35 infants in Studies 1C and 2C. Again, the higher maximum in Studies 1B and 2B reflects the two conditions. We did not specify in the preregistration which Bayes factor would govern the stopping rule—the first-look analysis, the looking-duration analysis, or both. In practice, we stopped when either measure reached a Bayes Factor of 8, or in the B studies when the comparison across conditions reached the threshold.

There were two deviations from the preregistrations regarding data collection. First, after testing 36 infants in Study 2A, we had only obtained a Bayes factor of 7. Although we met the target sample size, the Bayes factor was slightly below the target of 8 that we had set. We therefore considered increasing our sample size to better understand the effect in Study 2A. Optional stopping does not bias Bayesian inference (Rouder, 2014; Schönbrodt et al., 2017). Thus, we amended our preregistration to collect 10 additional infants in Study 2A, yielding a final sample of 46. Second, for Studies 1C and 2C, logistical constraints (i.e., the experimenter went on leave and then left the lab) led to a sample of 26 infants; nevertheless, the sample size is comparable to that of Thomas et al. (2022).

We used R (R Development Core Team, 2020) and JASP (Love et al., 2019) to calculate Bayes factors for the variables described above. We preregistered analyzing infants' anticipatory looking trials as a proportion of looking. We preregistered analyses of first looks only in Studies 1C and 2C; because Thomas et al. (2022) treated first looks as a primary dependent variable, we planned to analyze infants' first looks for Studies

1A, 1B, 2A, and 2B. To test whether infants looked first more often toward the thick relationship partner, we performed Bayesian binomial tests. To test whether infants spent more time looking at the thick relationship partner relative to the other actor, we had preregistered Bayesian *t*-tests; however, looking time distributions were not normal, violating an assumption of the *t*-test so we used Wilcoxon tests. The *t*-test results, reported in the supplementary materials, agree with the Wilcoxon results. For Studies 1B and 2B, which each included a test and control condition, we preregistered a two-sample *t*-test comparing the proportion of looking across conditions; owing to the same normality violation, we used two-sample Wilcoxon tests instead. Following Thomas et al. (2022), we also report comparisons of first looks across conditions in Studies 1B and 2B using contingency table tests (Bayesian chi-squared tests).

3. Study 1A

We examined whether infants anticipate that an actor who had shared saliva with a puppet would respond to the puppet's distress (Fig. 3).

3.1. Study 1A results

When the puppet expressed distress, the infants looked first (33/42, $BF_{10} = 195.52$) and longer (mean proportion = 0.62; $BF_{10} = 17.15$) at the actor who had shared food compared to the actor who had passed a ball with the puppet (Fig. 2). Thus, infants anticipated that the food-sharer would respond to the puppet's distress. Findings did not differ by age across studies (see supplemental materials).

4. Study 1B

On seeing two individuals share an orange slice, infants might have inferred either that the two had a thick relationship and respond to each other's distress, or that one individual is prosocial and responds to anyone's distress. Study 1B tested whether infants' predictions were specific to individuals who share saliva. In the control condition, the puppet from the familiarization videos was replaced with a different puppet in the anticipatory test trial (Fig. 2B).

4.1. Study 1B results

In the test condition, when the puppet expressed distress, the infants looked first (20/27, $BF_{10} = 5.97$) and longer (mean proportion = 0.67, $BF_{10} = 9.48$) at the actor who had shared food relative to the actor who had passed a ball (Fig. 2). In the control condition, when the puppet in the anticipatory looking trial was different, the infants did not look first (8/25, $BF_{01} = 0.60$) nor longer (mean proportion = 0.39, $BF_{01} = 1.23$) at the actor who had shared food. The pattern differed by condition for both the first looks ($BF_{10} = 62.24$) and proportion of looking ($BF_{10} = 8.48$).

5. Study 1C

We tested whether infants infer that individuals who share saliva respond specifically to each other's distress or instead to each other's behavior more broadly. In Study 1C's anticipatory test trial, the puppet requested a ball (Fig. 2B).

5.1. Study 1C Results

When the puppet requested a ball, the infants did not look first (12/25; $BF_{01} = 2.72$) nor longer (mean proportion = 0.55; $BF_{01} = 3.44$) to the actor who had shared food compared to the actor who had shared a ball (Fig. 2).

6. Study 1 discussion

After observing an actor place a single orange slice in both her mouth and a puppet's mouth, infants predicted that the actor, rather than the one who passed a ball with the puppet, would respond to the puppet's distress: They looked first and longer at the food-sharer when the puppet expressed distress. They did not do so when a different puppet expressed distress, nor when the original puppet requested a ball. These findings show that observing two actors engage in saliva sharing leads infants to predict those actors will respond to one another's distress, suggesting that the infants inferred a thick relationship.

Study 2 addressed three open questions. First, do infants view saliva sharing on its own as a cue to a thick relationship, or must food be present? In Study 1, the saliva-sharing interaction involved food, and infants may have tracked food sharing, rather than saliva sharing. Food sharing occurs in close relationships (Fiske, 1992), and children treat it as evidence of a relationship (Lieberman et al., 2017). Second, do infants infer that both individuals in a saliva-sharing interaction will be responsive to one another's distress, or do they only make such inferences of the actor who initiated the saliva-sharing interactions? In Study 1, the actor who initiated the saliva-sharing interactions was in the position to respond to the central actor's distress. Finally, what is the role of familiarity with an action in infants' inferences? Infants have likely become familiar with food sharing from their observations of others interacting in everyday life.

To address these questions, we presented infants with a saliva-sharing interaction involving no food: an actor put her finger in her own mouth, then in a puppet's mouth, and then in her own mouth again. Later, the actor expressed distress. We reasoned that infants may have less experience with actions involving fingers and mouths than they do with food-sharing actions. Critically, no food was involved in this saliva-sharing interaction, the actor was the one to initiate the saliva-sharing interaction, and the saliva-sharing interaction involved a relatively unfamiliar event. When the actor expressed distress, would infants nevertheless expect a response from the puppet who had shared saliva with her?

7. Study 2A

Here, an actor placed her finger in both a puppet's mouth and her own mouth, and placed her finger on both a second puppet's forehead and her own (Fig. 3). Neither action involved food, and only one action involved saliva sharing. If infants infer thick relationships from saliva sharing, then they should expect the puppet from the finger-in-mouths interaction to respond to the actor's distress.

7.1. Study 2A Results

When the actor expressed distress, the infants looked longer (mean proportion = 0.68; $BF_{10} = 174.9$) at the puppet from the finger-in-mouths interaction than to the puppet from the finger-on-foreheads interaction (Fig. 3). Evidence that infants looked first to the mouth puppet was inconclusive (22/34; $BF_{10} = 1.36$), consistent with the behavior of 8- to 10-month-olds in prior work (Thomas et al., 2022).

8. Study 2B

We examined whether infants' inferences about responsiveness to distress are specific to individuals who have shared saliva. In Study 2B's control condition, the actor from the familiarization videos was replaced with a different actor in the anticipatory test trial (Fig. 3B).

8.1. Study 2B results

In the test condition, when the actor from the familiarization trials expressed distress, the infants looked longer (mean proportion = 0.65,

$BF_{10} = 12.30$) at the puppet from the finger-in-mouths interaction than at the puppet with the forehead interaction (Fig. 3); evidence from first looks was inconclusive ($19/29$; $BF_{10} = 1.33$). In the control condition, when a novel actor expressed distress, infants did not look first ($14/28$; $BF_{01} = 2.38$) nor longer (mean proportion = 0.47, $BF_{01} = 4.76$) at the puppet who had engaged in the finger-in-mouths interaction. We found inconclusive evidence as to whether infants behaved differently across conditions (first looks $BF_{10} = 1.20$; proportion of looking $BF_{01} = 1.01$).

9. Study 2C

Do infants expect individuals in thick relationships to respond specifically to each other's distress, or generally to each other's communications and actions? To distinguish these possibilities, we presented infants with the same saliva-sharing events as in Studies 2A and 2B, followed by a new test event in which the actor expressed no distress and instead emitted a neutral vocalization that elicited the attention of the two puppets: "Whoop whoop" (Fig. 3B).

9.1. Study 2C results

When the actor said, "Whoop whoop", evidence for first looks was inconclusive ($13/22$; $BF_{01} = 1.63$), but there was evidence that the infants did not look did not look longer at the puppet who had participated in the saliva-sharing finger-in-mouths interaction (mean proportion = 0.54; $BF_{01} = 3.44$) than at the mouth puppet who had participated in the control, finger-on-foreheads interaction (Fig. 3).

10. Study 2 discussion

In Study 2, even when food was absent from the saliva-sharing interactions, infants reliably looked first toward the thick relationship partner in the test trials; infants predicted individuals who shared saliva would respond to each other's distress. Infants did not make such predictions when a different actor expressed distress, nor when the original actor behaved neutrally. Although infants looked longer toward the thick relationship partner in the test trials, they did not look first to the thick relationship partner. These findings replicate Thomas et al. (2022), who found that 8- to 10-month-old infants, in the same paradigm, looked longer, but not first, toward thick relationship partners. Infants' amount of looking may be a more sensitive measure of their expectations than are their first looks (refer to Ciftci et al., 2025).

11. Pooled analyses across papers on Infants' and Toddlers' understanding of saliva sharing

Across Studies 1 and 2, 4- to 6-month-old infants behaved similarly as the 8- to 10-month-old infants and 15- to 19-month-old toddlers previously tested. However, it is unclear whether infants' inferences about saliva sharing change with experience. To investigate this question, we pooled data across the current studies, Thomas et al.'s (2022) studies with 8- to 10-month-old infants and 15- to 19-month-old toddlers, and Ciftci et al.'s (2025) replication with 8- to 10-month-old infants. Relative to a 4-month-old infant, a 19-month-old infant has 4.75 times more lived experiences, and likely has much more experience participating in and observing saliva-sharing interactions. If experience shapes infants' and toddlers' understanding of saliva-sharing interactions, then effects should be stronger at older ages.

In exploratory analyses, we combined the data from all test conditions (total $n = 514$) across the present paper, Thomas et al. (2022), and Ciftci et al. (2025) and fit two Bayesian models with default priors: one for participants' first looks, and one for the proportion of time participants looked to the thick relationship partner. For each effect, we report the posterior median (PM; the Bayesian equivalent of a regression coefficient); the 95% credible interval (CI; the range includes the upper and lower bounds of 95% of the posterior distribution); and a Bayes

factor (BF; indicating the strength of the evidence in favor of the null hypothesis or the alternative hypothesis). We interpreted a finding as evidence for the alternative hypothesis when 0 was not in the 95% CI, and we used the BF to interpret the strength of the evidence, with numbers closer to 1 indicating inconclusive evidence.

11.1. First looks to the thick relationship partner

The first model was a logistic regression, appropriate for our binary dependent variable: whether participants looked first to the thick relationship partner during the test trial (Fig. 4A). The predictors were age (in months), the saliva-sharing paradigm (food-sharing in Study 1, or the finger-in-mouths action in Study 2), and the resulting statistical interaction. The fixed effects were centered. There was no evidence for an interaction (PM = 0.04, 95% CI [-0.03, 0.11], $BF_{01} = 12.5$, strong evidence in favor of the null), and there was mixed evidence that first looks differed depending on the paradigm (PM = -0.43, 95% CI [-0.83, -0.04], $BF_{01} = 2.36$, anecdotal evidence in favor of the null); although 0 was not in the 95% CI for the effect of paradigm, the Bayes factor being in favor of the null suggests inconclusive evidence. When we examined the intercept, we found that, as a group, participants looked first at the thick relationship partner (PM = 0.72, 95% CI [0.53, 0.92], $BF_{10} > 100$, decisive evidence in favor of the alternative). There was also evidence for a main effect of age (PM = 0.05, 95% CI [0.02, 0.19], $BF_{10} = 2.50$ in favor of the alternative): With every month of age, participants were about 5% more likely to look to the thick relationship partner when the central actor was in distress.

Although the interaction did not reach significance, the visual inspection of the data suggested that the slope of age was larger for the finger-in-mouths paradigm than for the food-sharing paradigm. We therefore further examined the effect of age on participants' first looks through analyses on the estimated marginal means (using the emmeans package in R; Lenth, 2016). This enabled us to separately examine the effect of age by paradigm. With age, participants were more likely to look first to the thick relationship partner for the finger-in-mouths paradigm (PM = 0.07, 95% CI [0.02, 0.12], $BF_{10} = 4.15$), but there was no effect of age on participants' likelihood of looking first to the thick relationship partner for the food-sharing action (PM = 0.03, 95% CI [-0.02, 0.08], $BF_{01} = 11.36$, strong evidence in favor of the null). The interaction between age and paradigm did not reach significance, so we interpret these analyses on the estimated marginal means with caution.

11.2. Proportion of looking at the thick relationship partner

To analyze age effects on the proportion of looking, we used a beta regression, appropriate for the data that are proportions between 0 and 1. The dependent variable was the proportion of time that infants looked to the thick relationship partner during the test trial (Fig. 4B). Beta regressions require that data be between 0 and 1, so we applied a transformation to limit the values, following Smithsen and Verkuilen (2006). Whereas the range of the original data was 0 to 1, the range of the transformed data was 0.0009 to 0.9990. The predictors were the same as that of the model on first looks. There was no evidence for an interaction (PM = -0.007, 95% CI [-0.02, 0.00], $BF_{01} = 66.66$, strong evidence in favor of the null), there was no evidence that the proportion of looking at the thick relationship partner differed depending on the paradigm (PM = 0.04, 95% CI [-0.04, 0.10], $BF_{01} = 41.66$, strong evidence in favor of the null), and there was no evidence that the proportion of looking at the thick relationship partner differed depending on age (PM = 0.006, 95% CI [0.00, 0.01], $BF_{01} = 15.87$, strong evidence in favor of the null). When we examined the intercept, we found that, as a group, participants looked longer to the thick relationship partner (PM = -0.38, 95% CI [-0.41, -0.34], $BF_{10} > 100$, decisive evidence in favor of the alternative).

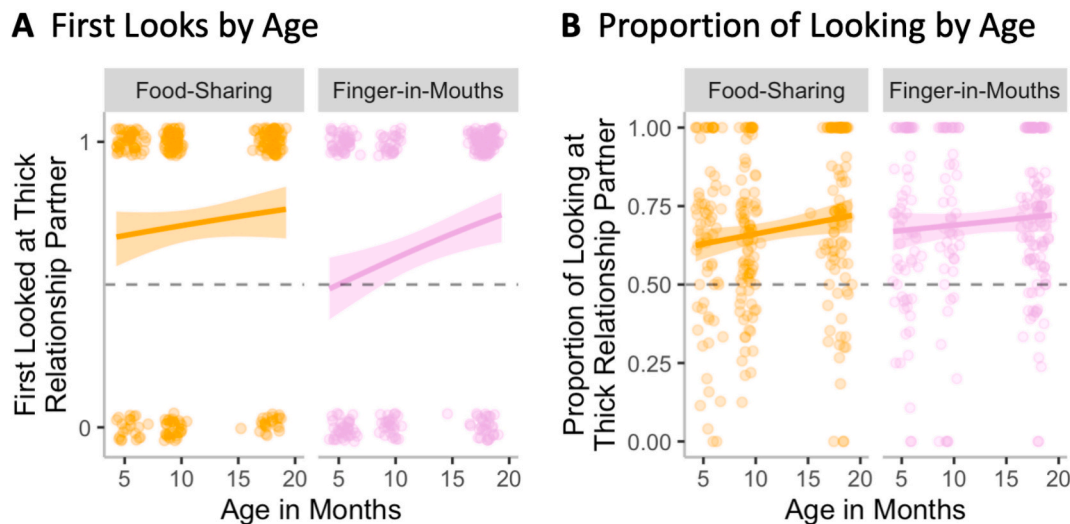


Fig. 4. Infants' and toddlers' (A) first looks and (B) proportion of looking at the thick relationship partner by age and by paradigm. The data (total $n = 514$) come from the present paper, from Thomas et al. (2022), and from Ciftci et al. (2025). The data in A are vertically jittered to more easily visualize patterns in the data.

11.3. Discussion of pooled age analyses

Despite a wide age range, we found little evidence that age predicted infants' responses. Only one of four measures showed evidence of an age effect: In the finger-in-mouths paradigm, older infants were more likely to look first toward the thick relationship partner. Age did not affect first looks or proportion of looking time in the food-sharing paradigm, nor did it affect proportion of looking time in the finger-in-mouths paradigm.

The near absence of age effects is striking. Participants ranged from 4 to 19 months—a 15-month span in which the oldest toddlers had roughly 4.75 times the life experience of the youngest infants. As noted above, the actual difference in experience is likely even larger than 4.75 times, given differences in amounts of sleep, visual acuity, the ability to sit up and locomote, and diet between younger infants and older toddlers (Barrera et al., 2018; Dias et al., 2018; Dobson & Teller, 1978; Frank et al., 2013; Iglowstein et al., 2003; Meek et al., 2022). Despite these differences in experience, only one of four measures showed evidence that participants' inferences changed with age.

Our findings suggest that knowledge of thick relationships is present and stable across the age range in the pooled analyses, from 4 to 19 months of age. This speaks to the role of experience in infants' and toddlers' inferences about thick relationships. Associative learning and empiricist accounts broadly (e.g., Benton & Hoch, 2026) predict that inferences should strengthen with more experience. Yet age did not predict the proportion of looking at the thick relationship partner in either paradigm, and the one age effect we did find, first looks in the finger-in-mouths paradigm, involves a behavior that is likely rarer in infants' everyday lives.

12. General discussion

Across two sets of studies, 4- to 6-month-old infants who observed an individual express distress looked toward the person with whom that individual had previously shared saliva, as though anticipating they would respond. They displayed this behavior both when they observed an interaction that involved food and an interaction that did not involve food, indicating that infants' inferences were driven by displays of physical intimacy rather than resource sharing. Infants had the same expectations when the individual in distress was the recipient of a saliva-sharing action (Study 1) or had initiated the saliva-sharing interaction (Study 2), indicating that infants anticipated mutual responsiveness. Infants did not have the same expectations (i) when a novel individual

expressed distress; nor (ii) when a familiar individual (who had been involved in the saliva-sharing interaction) requested a ball or vocalized without being distressed. Together, these results indicate that 4- to 6-month-old infants infer thick relationships from observed saliva-sharing interactions and use that inference to predict who will respond to distress.

This work extends a growing literature on how humans come to represent social relationships. Prior research shows that infants reason about dominance (Mascaro & Csibra, 2012; Thomas & Sarnecka, 2019; Thomsen et al., 2011) as well as affiliation and cooperation (Powell & Spelke, 2018; Tatone et al., 2015; Woo et al., 2023). Together with Thomas et al. (2022), our findings indicate that infants also differentiate affiliative relationships—differentiating thick from thin.

Our control experiments specify details about the underlying representations. First, infants did not expect the saliva-sharing partner to respond to an individual who had not previously been involved in the interaction. This suggests that their predictions were based on relationship representations constrained to specific people, in contrast to disposition representations, which would predict behavior directed at most or all people. Second, infants did not expect the saliva-sharing partner to respond when their partner asked for a ball or made a nonsense sound. This suggests that infants inferred a specific *relationship* between the individuals who share saliva that predicts a constrained set of behaviors between those two individuals, rather than a *trait* that predicts responsiveness generally.

The current findings constrain accounts of the role of experience in early developing relationship representations. Infants distinguished relationship types with four to six months of experience. They did so not only for food sharing, a social interaction that is likely highly familiar, but also for a less familiar action, in which an actor placed her finger in her own and then another's mouth.

Connectionist accounts propose that infants come to associate saliva sharing with responses to distress through experience (Benton & Hoch, 2026). Our results show that these inferences are already present very early, placing an upper bound on how much experience they require. More broadly, the early emergence of these inferences is consistent with an inductive bias rather than unconstrained, bottom-up associative learning, and with the possibility that infants possess core knowledge of social relationships that they draw on when interpreting these interactions.

The inductive bias account is further supported by exploratory analyses that pooled data from infants and toddlers, spanning 4 to 19 months of age, a roughly 375% increase in possible experience.

Measured behavior changed little across this period. Proportion of looking at the thick relationship partner did not vary with age in both the saliva-sharing and the finger-in-mouths paradigm, and the only age effect was an increase in first looks in the finger-in-mouths paradigm, which featured a behavior that is likely less common in infants' everyday lives compared with food sharing. Connectionist models predict the opposite pattern: Developmental change should be largest where experience with the behavior is greatest. Instead, these inferences are largely stable across development, with the only detectable change appearing where relevant experience is less. This provides further evidence against an account in which these inferences depend on associating behaviors through large amounts of experience and further constrains how much experience they require. Whether the amount of experience we find is sufficient (4 to 6 months) is compatible with connectionist accounts is an open question—one that would require knowing both the experience connectionist models need and how much experience infants actually have observing or experiencing these behaviors.

12.1. Future directions

The present findings raise multiple questions for future research. First, when an individual was in distress in the present paradigms, why did infants reliably look longer, but not first, at the thick relationship partner? Thomas et al. (2022) and Ciftci et al. (2025) reported similar patterns of findings: Thomas et al. (2022) in the finger-in-mouths paradigm (as in the present paper) and Ciftci et al. (2025) on the food-sharing paradigm. Our pooled analyses provided evidence of this pattern across the papers. We note, however, that toddlers successfully looked first at the thick relationship partner in both the food-sharing and finger-in-mouths paradigms of Thomas et al. (2022). The overall pattern may reflect that first looks are less sensitive than the proportion of looking (for discussion, see Ciftci et al., 2025). This may be due to it being a single measurement from a single timepoint as opposed to a measurement over seconds, or that making a predictive look first may be cognitively demanding as the proportion of looking measure allows participants more time to form a prediction.

Second, to what extent and how does experience shape infants' inferences about thick relationships? The present speaks to the effects of experience on the emergence of these inferences by studying 4- to 6-month-old infants, who have less experience observing and participating in acts of saliva sharing than do older infants and toddlers, and by testing the effects of viewing both common and less common saliva-sharing interactions. That said, 4- to 6-month-old infants' experience with saliva sharing is likely variable, even within our relatively homogenous sample of US American infants. Although the American Academy of Pediatrics recommends delaying solid foods until after 6 months, some infants begin to ingest solid foods prior to 4 months of age (Barrera et al., 2018; Dharod et al., 2023; Meek et al., 2022). Thus, many of the infants in our study likely had first-hand experience eating solid foods, and they may even have experienced sharing foods with others. Testing even younger infants or collecting longitudinal data could help shed light on how experience shapes infants' understanding and subsequent predictions.

Third, what form do these early representations take? These representations may respond to specific cues from their formation: because sharing bodily fluids occurs between kin across mammalian species, and marks thick relationships across cultures, humans may have evolved to treat it as evidence of thick relationships. Alternatively, these representations may start out as relatively abstract: Infants may hold inductive biases that direct their attention to how people in thick relationships behave and generalize from such behaviors, allowing them to quickly learn that saliva sharing occurs within thick relationships. On the

abstract account, these biases would also shape which actions are easiest to learn—and, because more learnable actions are more likely to be culturally adopted and transmitted, which actions recur across cultures (Samani & Thomas, 2026; Sperber & Hirschfeld, 2004).

Under either account, the capacity to reason about relationships may be a cognitive primitive, shaped by evolutionary pressures to navigate the social world (Seyfarth & Cheney, 2012; Thomas, 2026; Thomsen & Carey, 2013). Such a primitive could scaffold infants' learning about relationships within their social networks. We have focused on saliva sharing, but future studies could explore infants' sensitivity to other common markers of thick relationships, including other forms of physical intimacy, such as affective touch, or emotional intimacy such as comfort with distressed social partners (Dunbar, 2010; Sorokowska et al., 2021; Suvilehto et al., 2019; Woo et al., 2025), as well as how infants and children learn about markers that are culturally specific (e.g., taking off a head covering in some cultures can mean you have a very close social relationship with someone, while taking off a head covering in other cultures may just mean you are hot).

Fourth, how do these expectations compare to adult conceptions of close relationships? Intimacy has long been theorized as a distinct property of relationships—one not reducible to a relationship's strength or importance. Very formal relationships can be conceived of as strong, enduring, and valuable. In Fiske's (1992) relational-models framework, communal sharing construes close partners as sharing a common substance and body, a construal often enacted through commensality (eating together) and the exchange of bodily fluids (refer to Carsten, 1995; Clark & Mills, 1979). A complementary tradition in anthropology and sociolinguistics characterizes intimacy by reduced restraint: Thick relationship partners relax the barriers of politeness and formality that govern more distant ties, licensing behaviors such as candid disclosure, teasing, or close physical contact, that would be inappropriate among acquaintances (Brown & Levinson, 1987; Radcliffe-Brown, 1949).

Saliva sharing fits both accounts: It crosses a bodily boundary ordinarily maintained with all but one's closest partners. What remains open is whether infants' expectations track the merging of persons, the lowering of interpersonal barriers or both. Moreover, it remains unclear how infants construe social interactions (e.g. acts of social support) more broadly within thick relationships, whether infants expect thick relationships to endure over time, and how these representations may allow infants to learn about larger, interrelated social networks (e.g., families).

While many questions remain, our studies suggest that 4- to 6-month-old infants use saliva sharing to infer thick relationships and make predictions about who will respond to distress. An early sensitivity to intimate actions may allow infants to gauge social intimacy well before explicit instruction is possible, providing a foothold for learning culturally specific ways that intimacy is expressed and more broadly the structure and details of their own social environments.

CRedit authorship contribution statement

Brandon M. Woo: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Formal analysis, Data curation. **Elizabeth S. Spelke:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Rebecca Saxe:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Ashley J. Thomas:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest

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Appendix A. Supplementary data

Supplementary materials for this article can be found online at <https://doi.org/10.1016/j.evolhumbehav.2026.106960>.

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